



Problem Statement: MedConnect - Real-Time Local Healthcare Stock Locator

Event: Nexora Hackathon **Theme:** Google Technologies for Community Health & Smart Cities
Difficulty: Moderate (Progressive Complexity)



Background: The Real-World Problem

During medical emergencies, time is the most critical resource. When a patient needs an urgent, specific medication, specialized equipment (like an oxygen cylinder), or a rare blood group, families often find themselves running or calling across multiple local pharmacies and hospitals in Nashik.

Currently, there is no centralized, real-time inventory of what independent medical stores have in stock. Commercial delivery apps take hours or days, which is useless in an emergency. The lack of a hyperlocal, real-time tracking system leads to severe stress, wasted time, and potential loss of life.



The Challenge

Your task is to build a web-based, real-time emergency healthcare locator. The platform must connect local citizens looking for urgent medical supplies directly with nearby independent pharmacies and blood banks that currently have those items in stock.

The solution must be accessible, fast, and easy for non-technical pharmacy owners to update.



Requirements & Milestones

Level 1: The Baseline (Minimum Expected for Beginners)

To successfully submit this project, your team must build a functional web application with the following core features:

1. **User Portals:** A dual-interface system. One for "Citizens" to search for supplies, and one for "Vendors/Pharmacies" to update their inventory.
2. **Search & Filter:** A clean UI where users can search for a specific medicine by name and filter by availability.
3. **Basic Database CRUD:** Vendors must be able to Create, Read, Update, and Delete items from their stock list in real-time.

4. **Authentication:** Secure login for vendors to ensure inventory data cannot be tampered with by the public.

Level 2: The High Ceiling (For Advanced Teams to Win)

Teams aiming for the top prizes should implement the following advanced integrations to solve the real-life bottlenecks of this idea:

1. **AI Prescription Scanner:** Integrate the **Gemini API (Multimodal)**. Instead of typing complex medical names, a user can upload a photo of a doctor's handwritten prescription. The AI should extract the medicine names and automatically run the local search.
2. **Smart Proximity Routing:** Use the **Google Maps Platform (Distance Matrix API)**. Don't just list pharmacies alphabetically; rank them by real-time travel time, traffic conditions, and exact distance from the user's current location.
3. **Frictionless Vendor Updates:** Pharmacy owners are busy and might not log into a web app all day. Use **Google Cloud Functions** to build a lightweight SMS or WhatsApp integration where a vendor can simply text "Add 10 Paracetamol" to update the web database instantly.
4. **Real-Time Sync:** Utilize **Firebase Realtime Database** or **Firestore** so that when a pharmacy runs out of stock, the item disappears from the citizen's search screen instantly, without requiring a page refresh.



Recommended Tech Stack (Google Theme)

- **Frontend:** HTML/CSS/JavaScript (React, Vue, or Vanilla JS)
- **Backend / Database:** Firebase Authentication, Cloud Firestore, or Supabase
- **Hosting:** Firebase Hosting or Render
- **APIs:** Gemini API, Google Maps API, Google Cloud Vision